Hotel Booking System

Report 1: Project Description Report
UML Class Diagram Modelling Assignment | Instructor: Eric Maniraguha
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1. Project Title and Business Idea

Project title: Hotel Booking System.

Business idea: A hotel needs an information system to manage its clients, its rooms, the room reservations (bookings) made by clients, and the payments that settle those bookings.

2. Problem Statement and Purpose

Many small and medium hotels still manage reservations on paper or in scattered spreadsheets. This leads to recurring problems: double booking of the same room, lost or inconsistent client records, slow checking of room availability, unclear booking status (confirmed or cancelled), and weak tracking of payments and receipts.

Purpose of the proposed system: to centralise client, room, booking and payment information so that the hotel can register clients, know which rooms are available, record and cancel bookings, and process payments with receipts in an organised and reliable way.

3. Project Scope

The project focuses on **designing** the system with a UML class diagram. Developing a fully functional application is not required.

Functions covered

- Client registration and client profile update.
- Recording room details (type, price per night) and managing room availability status.
- Creating and cancelling bookings with check-in and check-out dates.
- Processing payments for bookings and generating receipts.

Functions excluded

- Hotel staff, payroll and human-resource management.
- Housekeeping, maintenance and inventory management.
- Restaurant, spa and other additional hotel services.
- Integration with online payment gateways or third-party booking websites.
- Advanced reporting, analytics and pricing strategies (discounts, seasonal rates).
- Implementation of the software (user interface, database, deployment).

4. Main Classes and Their Responsibilities

Class	Responsibility
Client	Represents a guest who registers in the system and makes bookings.
Room	Represents a physical hotel room with its type, price and availability.
Booking	Represents a reservation of one room by one client for a period of stay.
Payment	Represents the payment made to settle a booking, and its receipt.

5. Brief Description of System Requirements

Functional requirements

- FR1: The system shall allow a client to register and to update his or her profile (name, email, phone).
- FR2: The system shall store each room with its type, price per night and availability.
- FR3: The system shall check the availability of a room and update its status.
- FR4: The system shall create a booking linking one client to one room for a check-in and check-out date.
- FR5: The system shall allow a booking to be cancelled and its status to be updated.
- FR6: The system shall process the payment of a booking and generate a receipt.

Business rules and assumptions

- A client may have zero or many bookings; every booking belongs to exactly one client.
- A booking reserves exactly one room; a room may be booked many times over time (on non-overlapping dates).
- A booking is settled by a single payment (no instalments); a booking can exist before being paid, for example when it is cancelled before payment.

Non-functional requirements

- Usability: simple and clear screens for hotel receptionists.
- Reliability: no double booking of a room for the same dates.
- Data integrity: every booking and payment must be linked to valid records.